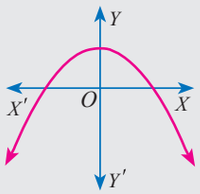
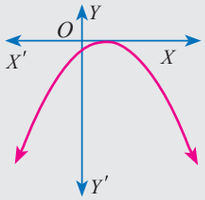
| | | | |
|---|---|---|---|
| 5. |  | 0 | Real and unequal roots |
| 6. |  | 1 | Real and unequal roots |

### 3.8.2 Solving quadratic equations through intersection of lines

We can determine roots of a quadratic equation graphically by choosing appropriate parabola and intersecting it with a desired straight line.

(i) If the straight line intersects the parabola at two distinct points, then the $x$ coordinates of those points will be the roots of the given quadratic equation.

(ii) If the straight line just touch the parabola at only one point, then the $x$ coordinate of the common point will be the single root of the quadratic equation.

(iii) If the straight line doesn't intersect or touch the parabola then the quadratic equation will have no real roots.

**Example 3.52** Draw the graph of $y = 2x^2$ and hence solve $2x^2 - x - 6 = 0$

***Solution*** **Step 1:** Draw the graph of $y = 2x^2$ by preparing the table of values as below

| $x$ | −2 | −1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $y$ | 8 | 2 | 0 | 2 | 8 |

**Step 2:** To solve $2x^2 - x - 6 = 0$, subtract $2x^2 - x - 6 = 0$ from $y = 2x^2$

$$\begin{array}{rl} y = 2x^2 & \\ 0 = 2x^2 - x - 6 & (-) \\ \hline y = x + 6 & \end{array}$$

The equation $y = x + 6$ represents a straight line. Draw the graph of $y = x+6$ by forming table of values as below

| $x$ | −2 | −1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $y$ | 4 | 5 | 6 | 7 | 8 |

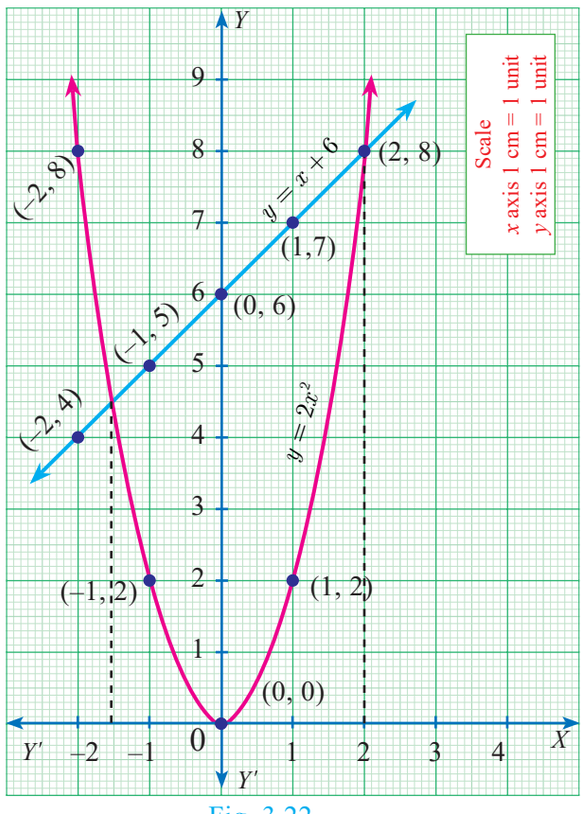



Fig. 3.22



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